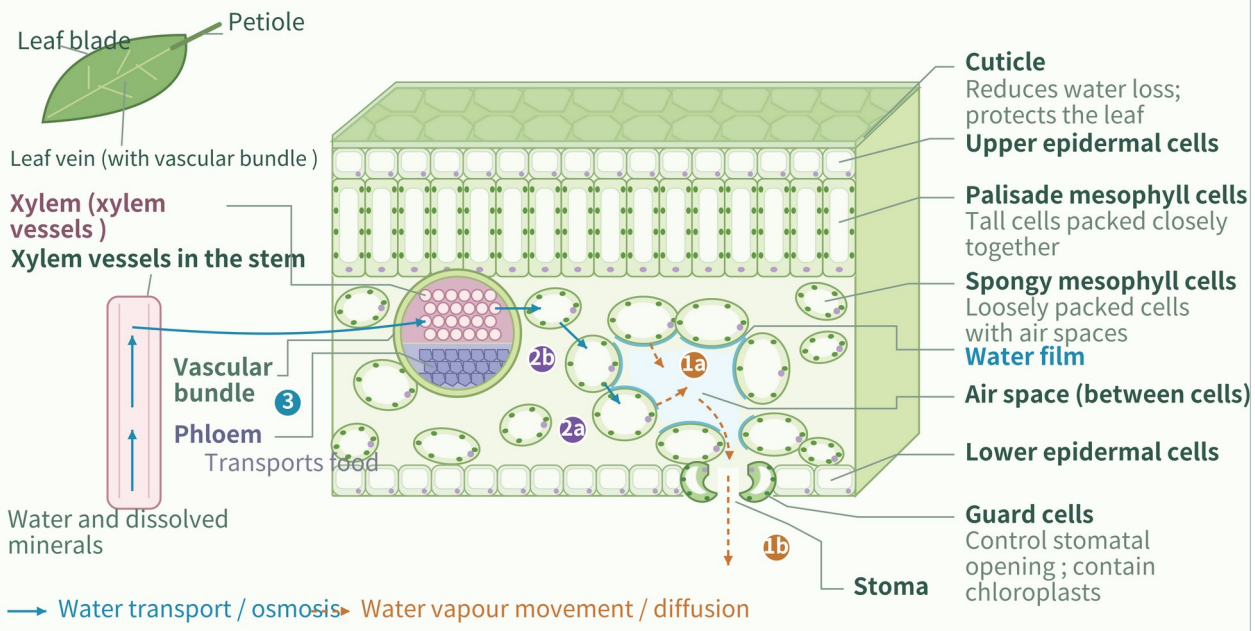
Worksheet 2: Transpiration and water transport

Name: _____ Class: _____ No.: _____

Topic notes

Internal leaf structure and transpiration

1: Evaporation / diffusion 2: Water loss / replacement 3: Transpiration pull



Transpiration through stomata

- 1a Water evaporates from the surfaces of mesophyll cells into the air spaces .
- 1b Water vapour in the air spaces diffuses through stomata into the air.

How transpiration pull forms

- 2a Mesophyll cells lose water, so their water potential falls.
- 2b Mesophyll cells take in water from nearby cells by osmosis.
- 3 Water is drawn out of the xylem vessels , creating transpiration pull .

1. Transpiration: water films evaporate; vapour diffuses through stomata.

2. Water replacement: mesophyll water potential falls; osmosis draws in water.

3. Upward transport: removing xylem water creates transpiration pull.

With adequate water and other conditions equal, stronger light favours **stomatal opening**.

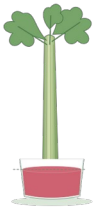
A celery stalk is a **petiole**.

Part 1 | Light intensity and water transport

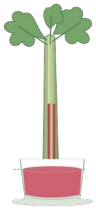
The gardening club asks: do light and leaves affect upward water transport? Light intensity is measured in lux.

Observation:

At the start

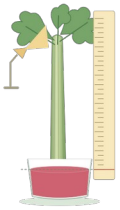


After some time



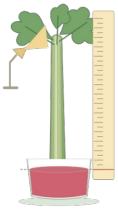
Experimental design:

Setup A • 10 cm



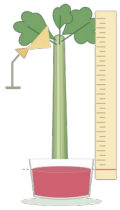
1200 lux

Setup B • 20 cm



300 lux

Setup C • 30 cm



133 lux

1. Identify variables: setup A, setup B and setup C

Tick the appropriate factors in each row. You may choose more than one per type.

Variable type	Factors
Independent variable the factor changed on purpose	☐ Light intensity (光強度) ☐ Height / average upward speed of the red water trace ☐ Temperature ☐ Humidity (濕度) and air movement ☐ Red water concentration and depth of immersion (浸入深度) ☐ Total leaf area of the setups with leaves ☐ Length and thickness of the celery stalks
Dependent variable the result measured	☐ Light intensity (光強度) ☐ Height / average upward speed of the red water trace ☐ Temperature ☐ Humidity (濕度) and air movement ☐ Red water concentration and depth of immersion (浸入深度) ☐ Total leaf area of the setups with leaves ☐ Length and thickness of the celery stalks
Controlled variables the factors kept the same	☐ Light intensity (光強度) ☐ Height / average upward speed of the red water trace ☐ Temperature ☐ Humidity (濕度) and air movement ☐ Red water concentration and depth of immersion (浸入深度) ☐ Total leaf area of the setups with leaves ☐ Length and thickness of the celery stalks

2. Select the assumptions for a fair comparison (select all that apply)

An assumption is a condition we have not proved but take as true, to help make the experiment results valid.

☐ Changing the lamp distance does not make the setups have different temperatures.

☐ Setups A, B and C have the same total leaf area.

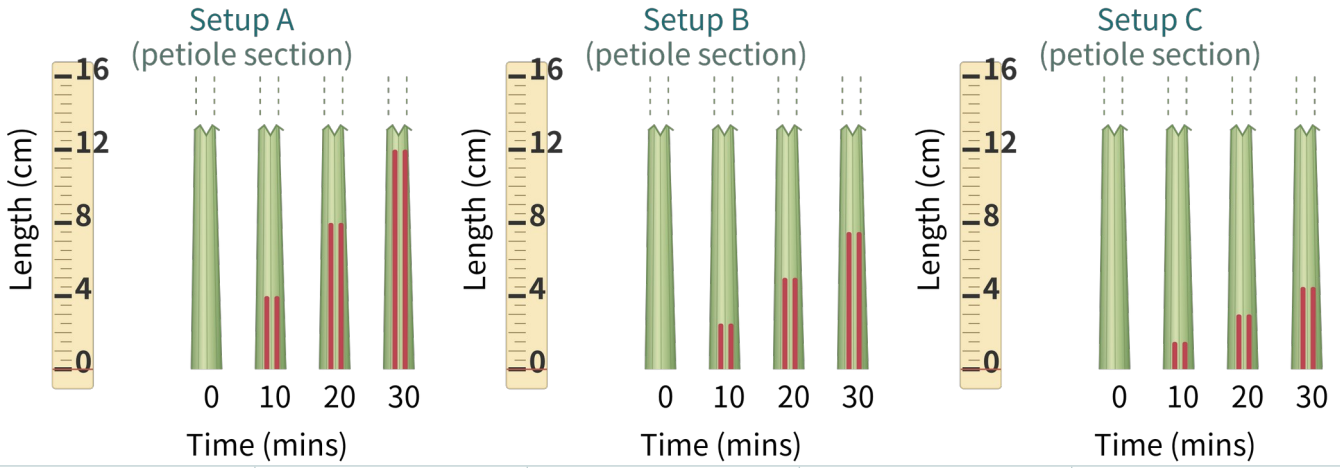
☐ Leaf area does not affect the results, so it does not need to be kept the same.

☐ Humidity (濕度) and air movement are the same for all setups.

3. Main investigation: setup A, setup B and setup C readings

Use the printed ruler marks; read heights to one decimal place (cm).

Red water-trace height changes in setup A, setup B and setup C



Time (mins)	0	10	20	30
Setup A				
Setup B				
Setup C				

Part 2 | Leaves and water transport

4. Does the presence of leaves affect water uptake and upward water transport?

Does this extension investigation need a control setup?

If so, how would you design the comparison?

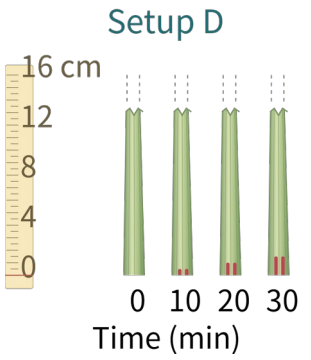
(Hint: a control setup provides a comparison baseline. Keep conditions other than the factor being investigated the same.)

(Hint: how is the control setup different from the experiment setup?)

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5. Extension readings

Reuse setup B from item 3. Read and record only leafless setup D (one decimal place, cm).



min	0	10	20	30
Setup D				

6. Average upward speed of the red-trace top

(30 min reading − 0 min reading) ÷ 30; give two decimal places (cm/min).

Setup A	Setup B	Setup C	Setup D

7. Light intensity and water-trace speed

Plot setup A, setup B, setup C and setup D from your calculations. Join setup A, setup B and setup C with a smooth trend line; keep setup D separate.

Setup A: 1200 lux Setup B: 300 lux Setup C: 133 lux Setup D: 300 lux



Part 2 | Results and analysis (continued)

8. How is light intensity (光強度) related to the upward speed of the water trace?

As light intensity (光強度) increases, the average upward speed of the red water trace usually will （increase / decrease / show no clear change / show no consistent trend）.

9. What is the role of leaves?

At the same 20 cm lamp distance, the red-trace speed in leafy setup B, compared with leafless setup D, is （faster / slower / similar）.

10. What is the role of leaves?

This comparison supports the idea that leaves （help water move upwards / stop water moving upwards / have no effect on water moving upwards）.

11. What can this experiment tell us?

We measured only the height of the red water trace. We did not measure water vapour (水汽) lost from the leaves. Which statement is most suitable?

☐ The red water trace indicates water uptake and upward water transport, but does not represent leaf transpiration rate (蒸騰速率) (the rate of water loss through stomata (氣孔)).

☐ The upward speed of the water trace is the rate of water vapour (水汽) loss from leaves and can be used directly as the transpiration rate (蒸騰速率).

☐ If the setups with leaves are faster than the setup without leaves, this fully proves that water vapour (水汽) is lost only through stomata (氣孔).

Part 3 | Integrated explanation

12. Using the experimental results and learning points, explain how light intensity (光強度) and leaves affect water transport, and explain how transpiration (蒸騰) and transpiration pull (蒸騰拉力) arise.